



# Graphs in Machine Learning

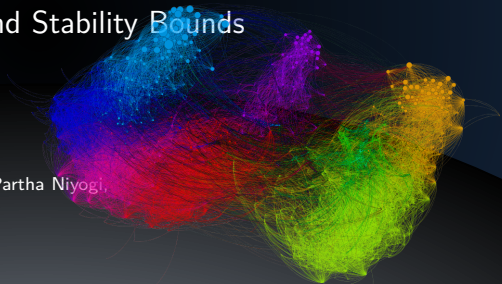
## SSL Regularization and Stability

Regularized, Soft Harmonic, and Stability Bounds

Michal Valko

*Isara Labs*

Partially based on material by: Mikhail Belkin, Partha Niyogi,  
Olivier Chapelle, Bernhard Schölkopf



# SSL with Graphs: Regularized Harmonic Functions

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...and therefore we simply add  $\gamma_G$  to the diagonal of  $\mathbf{L}$ !

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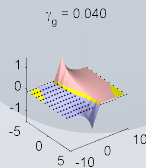
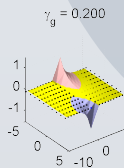
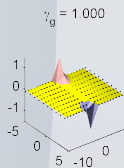
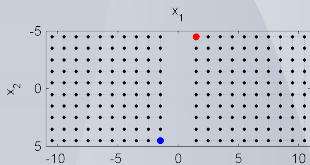
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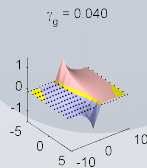
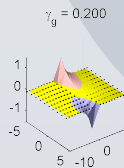
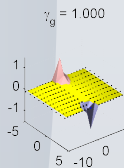
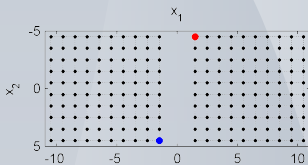


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What happens to sneaky outliers?

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$\mathbf{y} \equiv$  pseudo-targets with  $y_i = \begin{cases} \text{true label} & \text{for labeled examples} \\ 0 & \text{otherwise.} \end{cases}$

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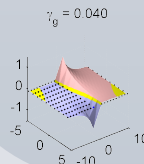
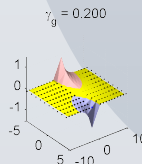
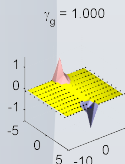
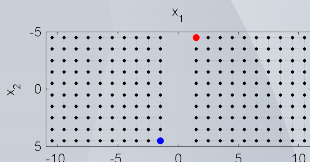
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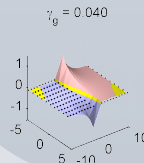
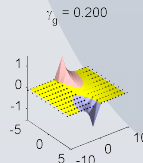
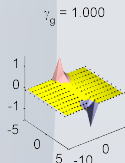
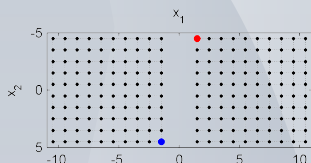


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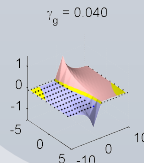
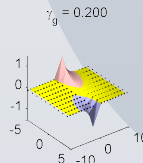
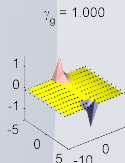
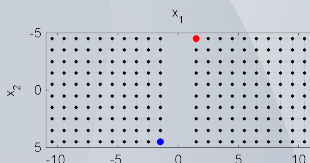
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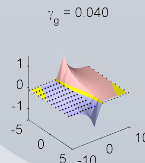
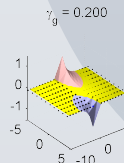
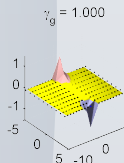
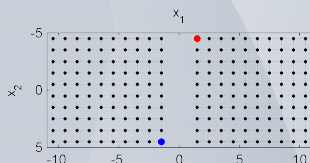
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Provable generalization guarantees for the soft one.

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$$\mathbf{f}_2^* - \mathbf{f}_1^* = \mathcal{C}_2^{-1} \mathbf{y}_2 - \mathcal{C}_1^{-1} \mathbf{y}_1$$

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$$\mathbf{f}^* = \min_{\mathbf{f} \in \mathbb{R}^N} (\mathbf{f} - \mathbf{y})^\top \mathbf{C} (\mathbf{f} - \mathbf{y}) + \mathbf{f}^\top \mathbf{Q} \mathbf{f}$$

Think about **stability** of this solution.

Consider two datasets differing in exactly one *labeled* point.

$$\mathcal{C}_1 = \mathbf{C}_1^{-1} \mathbf{Q} + \mathbf{I} \text{ and } \mathcal{C}_2 = \mathbf{C}_2^{-1} \mathbf{Q} + \mathbf{I}$$

What is the maximal difference in the solutions?

$$\begin{aligned} \mathbf{f}_2^* - \mathbf{f}_1^* &= \mathcal{C}_2^{-1} \mathbf{y}_2 - \mathcal{C}_1^{-1} \mathbf{y}_1 \\ &= \mathcal{C}_2^{-1} (\mathbf{y}_2 - \mathbf{y}_1) - (\mathcal{C}_1^{-1} - \mathcal{C}_2^{-1}) \mathbf{y}_1 \end{aligned}$$

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Note that  $\mathbf{v} \in \mathbb{R}^{N \times 1}$ ,  $\lambda_m(A) \|\mathbf{v}\|_2 \leq \|A\mathbf{v}\|_2 \leq \lambda_M(A) \|\mathbf{v}\|_2$

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# SSL with Graphs: Stability Bounds

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# SSL with Graphs: Stability Bounds

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# SSL with Graphs: Stability Bounds

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# SSL with Graphs: Stability Bounds

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Using  $\lambda_m(\mathcal{C}) \geq \frac{\lambda_m(\mathbf{Q})}{\lambda_M(\mathbf{C})} + 1$

# SSL with Graphs: Stability Bounds

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Using  $\lambda_m(\mathbf{C}) \geq \frac{\lambda_m(\mathbf{Q})}{\lambda_M(\mathbf{C})} + 1$

$$\|\mathbf{f}_2^* - \mathbf{f}_1^*\|_2 \leq \frac{\|\mathbf{y}_2 - \mathbf{y}_1\|_2}{\frac{\lambda_m(\mathbf{Q})}{\lambda_M(\mathbf{C}_1)} + 1} + \frac{\lambda_M(\mathbf{Q}) \|\mathbf{C}_1^{-1} - \mathbf{C}_2^{-1}\|_2 \cdot \|\mathbf{y}_1\|_2}{\left(\frac{\lambda_m(\mathbf{Q})}{\lambda_M(\mathbf{C}_2)} + 1\right) \left(\frac{\lambda_m(\mathbf{Q})}{\lambda_M(\mathbf{C}_1)} + 1\right)}$$

## SSL with Graphs: Stability Bounds

$$\|\mathbf{f}_2^* - \mathbf{f}_1^*\|_\infty \leq \beta \leq \frac{\|\mathbf{y}_2 - \mathbf{y}_1\|_2}{\frac{\lambda_m(\mathbf{Q})}{\lambda_M(\mathbf{C}_1)} + 1} + \frac{\lambda_M(\mathbf{Q}) \|\mathbf{C}_1^{-1} - \mathbf{C}_2^{-1}\|_2 \cdot \|\mathbf{y}_1\|_2}{\left(\frac{\lambda_m(\mathbf{Q})}{\lambda_M(\mathbf{C}_2)} + 1\right) \left(\frac{\lambda_m(\mathbf{Q})}{\lambda_M(\mathbf{C}_1)} + 1\right)}$$



## SSL with Graphs: Stability Bounds

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Now, let us plug in the values for our problem.

## SSL with Graphs: Stability Bounds

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Now, let us plug in the values for our problem.

Take  $c_l = 1$

## SSL with Graphs: Stability Bounds

$$\|\mathbf{f}_2^* - \mathbf{f}_1^*\|_\infty \leq \beta \leq \frac{\|\mathbf{y}_2 - \mathbf{y}_1\|_2}{\frac{\lambda_m(\mathbf{Q})}{\lambda_M(\mathbf{C}_1)} + 1} + \frac{\lambda_M(\mathbf{Q})\|\mathbf{C}_1^{-1} - \mathbf{C}_2^{-1}\|_2 \cdot \|\mathbf{y}_1\|_2}{\left(\frac{\lambda_m(\mathbf{Q})}{\lambda_M(\mathbf{C}_2)} + 1\right) \left(\frac{\lambda_m(\mathbf{Q})}{\lambda_M(\mathbf{C}_1)} + 1\right)}$$

Now, let us plug in the values for our problem.

Take  $c_l = 1$  and  $c_l > c_u$ .

## SSL with Graphs: Stability Bounds

$$\|\mathbf{f}_2^* - \mathbf{f}_1^*\|_\infty \leq \beta \leq \frac{\|\mathbf{y}_2 - \mathbf{y}_1\|_2}{\frac{\lambda_m(\mathbf{Q})}{\lambda_M(\mathbf{C}_1)} + 1} + \frac{\lambda_M(\mathbf{Q})\|\mathbf{C}_1^{-1} - \mathbf{C}_2^{-1}\|_2 \cdot \|\mathbf{y}_1\|_2}{\left(\frac{\lambda_m(\mathbf{Q})}{\lambda_M(\mathbf{C}_2)} + 1\right) \left(\frac{\lambda_m(\mathbf{Q})}{\lambda_M(\mathbf{C}_1)} + 1\right)}$$

Now, let us plug in the values for our problem.

Take  $c_l = 1$  and  $c_l > c_u$ . We have  $|y_i| \leq 1$

## SSL with Graphs: Stability Bounds

$$\|\mathbf{f}_2^* - \mathbf{f}_1^*\|_\infty \leq \beta \leq \frac{\|\mathbf{y}_2 - \mathbf{y}_1\|_2}{\frac{\lambda_m(\mathbf{Q})}{\lambda_M(\mathbf{C}_1)} + 1} + \frac{\lambda_M(\mathbf{Q})\|\mathbf{C}_1^{-1} - \mathbf{C}_2^{-1}\|_2 \cdot \|\mathbf{y}_1\|_2}{\left(\frac{\lambda_m(\mathbf{Q})}{\lambda_M(\mathbf{C}_2)} + 1\right) \left(\frac{\lambda_m(\mathbf{Q})}{\lambda_M(\mathbf{C}_1)} + 1\right)}$$

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## SSL with Graphs: Stability Bounds

$$\|\mathbf{f}_2^* - \mathbf{f}_1^*\|_\infty \leq \beta \leq \frac{\|\mathbf{y}_2 - \mathbf{y}_1\|_2}{\frac{\lambda_m(\mathbf{Q})}{\lambda_M(\mathbf{C}_1)} + 1} + \frac{\lambda_M(\mathbf{Q})\|\mathbf{C}_1^{-1} - \mathbf{C}_2^{-1}\|_2 \cdot \|\mathbf{y}_1\|_2}{\left(\frac{\lambda_m(\mathbf{Q})}{\lambda_M(\mathbf{C}_2)} + 1\right) \left(\frac{\lambda_m(\mathbf{Q})}{\lambda_M(\mathbf{C}_1)} + 1\right)}$$

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$$\beta \leq 2 \left[ \frac{\sqrt{2}}{\lambda_m(\mathbf{Q}) + 1} + \sqrt{2n_l} \frac{1 - c_u}{c_u} \frac{\lambda_M(\mathbf{Q})}{(\lambda_m(\mathbf{Q}) + 1)^2} \right]$$

## SSL with Graphs: Stability Bounds

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$\mathbf{Q}$  is reg.  $\mathbf{L}$ :  $\lambda_m(\mathbf{Q}) = \lambda_m(\mathbf{L}) + \gamma_g$

## SSL with Graphs: Stability Bounds

$$\|\mathbf{f}_2^* - \mathbf{f}_1^*\|_\infty \leq \beta \leq \frac{\|\mathbf{y}_2 - \mathbf{y}_1\|_2}{\frac{\lambda_m(\mathbf{Q})}{\lambda_M(\mathbf{C}_1)} + 1} + \frac{\lambda_M(\mathbf{Q})\|\mathbf{C}_1^{-1} - \mathbf{C}_2^{-1}\|_2 \cdot \|\mathbf{y}_1\|_2}{\left(\frac{\lambda_m(\mathbf{Q})}{\lambda_M(\mathbf{C}_2)} + 1\right) \left(\frac{\lambda_m(\mathbf{Q})}{\lambda_M(\mathbf{C}_1)} + 1\right)}$$

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## SSL with Graphs: Stability Bounds

$$\|\mathbf{f}_2^* - \mathbf{f}_1^*\|_\infty \leq \beta \leq \frac{\|\mathbf{y}_2 - \mathbf{y}_1\|_2}{\frac{\lambda_m(\mathbf{Q})}{\lambda_M(\mathbf{C}_1)} + 1} + \frac{\lambda_M(\mathbf{Q})\|\mathbf{C}_1^{-1} - \mathbf{C}_2^{-1}\|_2 \cdot \|\mathbf{y}_1\|_2}{\left(\frac{\lambda_m(\mathbf{Q})}{\lambda_M(\mathbf{C}_2)} + 1\right) \left(\frac{\lambda_m(\mathbf{Q})}{\lambda_M(\mathbf{C}_1)} + 1\right)}$$

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## SSL with Graphs: Stability Bounds

$$\|\mathbf{f}_2^* - \mathbf{f}_1^*\|_\infty \leq \beta \leq \frac{\|\mathbf{y}_2 - \mathbf{y}_1\|_2}{\frac{\lambda_m(\mathbf{Q})}{\lambda_M(\mathbf{C}_1)} + 1} + \frac{\lambda_M(\mathbf{Q}) \|\mathbf{C}_1^{-1} - \mathbf{C}_2^{-1}\|_2 \cdot \|\mathbf{y}_1\|_2}{\left(\frac{\lambda_m(\mathbf{Q})}{\lambda_M(\mathbf{C}_2)} + 1\right) \left(\frac{\lambda_m(\mathbf{Q})}{\lambda_M(\mathbf{C}_1)} + 1\right)}$$

Now, let us plug in the values for our problem.

Take  $c_l = 1$  and  $c_l > c_u$ . We have  $|y_i| \leq 1$  and  $|f_i^*| \leq 1$ .

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This algorithm is  $\beta$ -stable!

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**Larger implications of random walks**

random walk relates to **commute distance**

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Solution: **Manifold Regularization**

# Michal Valko

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Isara Labs



<https://misovalko.github.io/mva-ml-graphs.html>